

KONGU ENGINEERING COLLEGE · CSE & ECE

Smart Construction Site Safety Monitor

Review 1 — Complete Presentation Kit

Slide-by-Slide Script • Anticipated Jury Q&A • Case Study & Survey Evidence

Team: Jayasurya K · Harishraj V · Sunmathi S · Twishaa T

Mentor: Kiruthika A, AP/CSE | Co-Mentor: Dr. M. Pavithra, AP/ECE

Part A

Slide-by-Slide Presentation Script

One explanation per slide, mapped to your final 8-slide deck

SLIDE 01

Title Slide — Team & Project Introduction

Presenter: Harishraj V (Team Lead)

Good morning/afternoon. We are a team from CSE & ECE, presenting our Review 1 project "**Smart Construction Site Safety Monitor**" — an AI-powered Computer Vision and Wearable IoT system for real-time worker safety. Our team includes Jayasurya K, myself Harishraj V, Sunmathi S and Twishaa T, guided by Kiruthika A (AP/CSE) and Dr. M. Pavithra (AP/ECE). Over the next few slides we'll cover the problem, our proposed solution, the system workflow, and where we currently stand in implementation.

Key Points to Emphasize

- Say the project title clearly and slowly — first impression for the jury.
- Introduce every team member by name and roll number if asked.
- One line up front on what makes it different: AI camera detection + wearable IoT combined.
- Mention it's Review 1 so the jury calibrates expectations on depth of completion.

SLIDE 02

Problem Statement

Presenter: Jayasurya K

Construction sites are among the most hazardous workplaces globally. We identified six key gaps: workers frequently skip **PPE like helmets, vests, and gloves**; safety officers cannot manually supervise every worker at all times, making monitoring slow and error-prone; falls and hazards are usually noticed only **after** an accident occurs; invisible risks like CO and CH₄ gas go undetected without dedicated sensors; most existing systems only log incidents instead of alerting in real time; and advanced multi-camera AI safety systems remain too costly for small and medium construction firms.

Key Points to Emphasize

- Group the 6 problems: human-side gaps vs environment-side gaps vs cost gap.
- Emphasize "detected only after the accident" — this is the core motivation.
- If the jury asks for data here, this is backed by real statistics — see Part C of this document.
- Keep this under 90 seconds; specifics come in the solution slide.

SLIDE 03

Proposed Solution

Presenter: Sunmathi S

Our solution has two connected parts. The **Software Module** uses a mobile camera feed processed frame-by-frame with OpenCV and passed to our fine-tuned **YOLO11** model, which detects the person and flags missing helmet, vest, or other PPE instantly. The **Wearable Smart Helmet**, built on ESP32, adds a second safety layer: MPU6050 for fall and abnormal-movement detection, MQ-7 for Carbon Monoxide monitoring, and DHT22 for temperature and humidity. All data reaches a central ESP32 base station that drives the buzzer, LEDs, and OLED display for instant on-site warnings.

Key Points to Emphasize

- Split explanation clearly: software = "eyes" (camera + AI), hardware = "body sensors" (helmet).
- Say MPU6050 covers what a camera can't — a fall that happens off-camera or behind an obstruction.
- This is your core USP slide — combine vision + wearable IoT, keep energy high.
- If asked about the dashboard here, briefly note it's covered in the Implementation Status slide.

SLIDE 04

Model Training & Workflow

Presenter: Twishaa T

This slide covers two things — how we built the model, and how the system runs live. **Training** happened on Google Colab with GPU acceleration: we verified our datasets, loaded the pretrained **YOLO11n** weights, applied data augmentation for robustness, trained for **100 epochs**, validated using Precision, Recall and mAP, and exported the final weights as **best.pt** and **ONNX** for flexible deployment. **At runtime**, the pipeline is: Mobile Camera → YOLO11 Detection → Decision Logic → ESP32 Base Station → Alert Output. The base station also takes live sensor input — CO level from MQ-7, temperature/humidity from DHT22, fall detection from MPU6050, and manual SOS button presses — and responds with buzzer, red LED, OLED warning, and a dashboard log entry.

Key Points to Emphasize

- Treat this as two mini-stories: "how we trained it" then "how it runs live" — don't blur them.
- If asked why 100 epochs: standard convergence point for YOLO fine-tuning on a moderate dataset size.
- Mention best.pt vs ONNX only if asked: ONNX enables deployment outside PyTorch (edge/cross-platform).
- End on the output row — buzzer, LED, OLED, dashboard log — as the tangible result of both pipelines.

SLIDE 05

Existing vs Proposed System

Presenter: Jayasurya K

Existing AI-based PPE detection research typically uses CCTV cameras with YOLO-family models to detect helmets and vests, shown on a monitoring dashboard. These are accurate and fast, but only detect **visual** PPE violations — they cannot detect falls, cannot monitor environmental conditions like gas or temperature, and offer no wearable device or strong emergency alert mechanism. Our system keeps everything existing approaches do well and adds four new capabilities: fall detection, a wearable smart helmet, CO gas monitoring, and temperature monitoring — while staying low-cost by using a mobile camera instead of expensive CCTV infrastructure.

Key Points to Emphasize

- Use the comparison table as your script — read row by row, pause on every "No → Yes".
- Say clearly: "we are not replacing existing approaches, we are extending them."
- Fall detection and gas monitoring are your two strongest differentiators — emphasize both.
- This slide is backed by published research — see Part C, Section C3 of this kit.

SLIDE 06

Tech Stack & Components

Presenter: Sunmathi S

On the software side: **YOLO11** as our fine-tuned detection model, **OpenCV** for real-time frame processing, and **Python** for the decision logic connecting detection to hardware action. On the hardware side: the **ESP32** controls both the wearable helmet and the base station; **MPU6050** handles fall and

motion sensing; **MQ-7** detects carbon monoxide; **DHT22** covers temperature and humidity; and the **OLED display and buzzer** form our on-site visual and audio alert output.

Key Points to Emphasize

- Group by software vs hardware as you speak — mirrors the Proposed Solution slide structure.
- If asked why YOLO11: lightweight, fast inference, strong accuracy-speed trade-off for edge use.
- Mention best.pt / ONNX export briefly if asked about deployment flexibility.
- Keep this fast-paced — it's a reference slide, not a story slide.

SLIDE 07

Implementation Status

Presenter: Harishraj V

Our overall project completion stands at **30%**, with the project plan weighted 50% software and 50% hardware. **Completed:** two PPE datasets collected and annotated, the YOLO11 model trained and validated, live detection tested successfully on webcam, ESP32 wiring with buzzer/LED circuit completed, and the **dashboard UI designed** — currently running on demo data. **Pending:** integrating the MQ-7 and DHT22 sensors, linking the dashboard UI to live model output, mounting the full smart helmet prototype, and field testing on an actual site.

*Always describe the dashboard as "**designed and functional on demo/simulated data, not yet live-linked to the detection model**" — accurate and confident, never oversold.*

Key Points to Emphasize

- Be transparent about 30% — juries respect honesty over over-claiming progress.
- State plainly: "dashboard is fully designed and working on demo data — live linking to the model is a scoped, near-term task, not an unsolved problem."
- Group pending items as your "integration phase" — signals a clear next step, not a stuck project.
- Expect the most follow-up questions on this slide — see Part B, Category 4.

SLIDE 08

Thank You

Presenter: All Team Members

We thank the panel for their time and attention. This concludes our Review 1 presentation of the Smart Construction Site Safety Monitor — we're happy to take any questions.

Key Points to Emphasize

- Deliver as a group moment — all four members can stand together for this slide.
- Don't rush off — pause, make eye contact with the panel, invite questions confidently.
- Whoever answers a question should restate it briefly before answering — buys thinking time.
- If a question doesn't map to your slide, it's fine for any teammate to jump in.

Part B

Anticipated Jury Questions & Suggested Answers

Organized by category — read the question, deliver the answer in your own words

1 · AI / Computer Vision

Q: Why did you choose YOLO11 over YOLOv5, YOLOv8, or Faster R-CNN?

YOLO11 is the latest in the YOLO family, giving a strong balance of speed and accuracy needed for real-time inference on standard hardware, with better small-object handling than earlier versions. As a single-stage detector, it's also significantly faster than two-stage detectors like Faster R-CNN, which matters directly for our live-alert use case.

Q: What metrics did you use to evaluate the model, and how did it perform?

We used Precision, Recall, and mAP@50 — the standard object-detection metrics — generated during YOLO validation along with a confusion matrix, PR curve, and F1 curve. (Quote your actual training-run numbers here from your notebook results.)

Q: How do you handle false positives or missed detections — e.g., occluded PPE or odd camera angles?

Our training used data augmentation across varied angles, lighting, and partial occlusion to improve robustness, and we apply a 0.5 confidence threshold to filter weak detections. The alert cooldown also prevents a single misread frame from triggering a full hardware alarm.

Q: What's in your dataset — how many images, how many classes?

We combined two annotated PPE datasets covering classes like helmet, gloves, boots, and goggles along with their violation counterparts (no_helmet, no_gloves, etc.), verified for label-image pairing before training.

2 · Hardware / IoT

Q: Why ESP32 instead of Arduino or Raspberry Pi?

ESP32 offers built-in Wi-Fi/Bluetooth, enough GPIO pins for multiple sensors, low power draw, and low cost — ideal for a wearable form factor. Raspberry Pi is bulkier and power-hungry for a helmet-mounted unit, and basic Arduino lacks wireless connectivity we'll need for the deployable version.

Q: How does data move from the camera/PC to the ESP32, and why USB serial and not wireless?

Currently we use USB serial at 115200 baud for a reliable, low-latency link during the prototype and demo stage. Moving to wireless (Wi-Fi or BLE) is planned for the fully wearable, mobile version once the core detection-to-alert pipeline is proven.

Q: What's the power source and battery life for the wearable helmet?

We're currently validating sensors on a benchtop supply; battery integration (Li-ion cell with a charging circuit) is scoped as part of the full helmet-mounting task, listed under Pending on the Implementation Status slide.

Q: Why MQ-7 specifically for gas sensing, and not MQ-2 or MQ-135?

MQ-7 is optimized specifically for Carbon Monoxide, one of the most common and dangerous invisible hazards at construction and demolition sites — from generators, welding fumes, or enclosed excavation spaces. We chose targeted, reliable CO sensing over a general-purpose gas sensor.

3 · System Design & Integration

Q: What is the alert cooldown and why is it necessary?

The 5-second cooldown stops the system from firing repeated ALERT signals for the same ongoing violation across consecutive frames, which would otherwise flood the buzzer/LED and cause "alarm fatigue" — making warnings less meaningful to workers over time.

Q: The dashboard shows demo data — why isn't it linked to the live model yet?

At 30% completion, we prioritized validating the core detection pipeline and hardware alert loop first. The dashboard UI itself is fully designed and functional; linking it to live model output is a scoped, near-term integration task — not an unresolved design problem.

Q: How would this scale to multiple workers or multiple cameras across a large site?

The architecture is modular — each camera-and-model pair can independently feed the same base station/dashboard logic, so scaling means adding more camera-model units rather than redesigning the core pipeline. Cloud-based inference is a future option for larger multi-camera deployments.

Q: Isn't CCTV + YOLO PPE detection already a solved problem in existing research?

Yes — PPE detection via YOLO is a well-validated research area (see Part C, Section C3). Where we differ is combining that vision layer with a wearable IoT helmet for fall detection and gas/temperature sensing — hazards a camera alone can never catch, since it can't see behind an obstruction or detect an invisible gas.

4 · Practical / Project Management

Q: What's the approximate cost of your hardware setup?

(Fill in your actual component cost total here — ESP32, MPU6050, MQ-7, DHT22, buzzer, LEDs, OLED typically total a modest amount in India.) Frame it against expensive multi-camera industrial AI safety systems to reinforce your low-cost differentiator.

Q: Is worker video footage stored — any privacy concerns?

Currently the system performs on-the-fly detection without storing raw video; only violation events (timestamp and class) are logged for reporting. This can be explicitly designed around data-minimization principles as the system matures.

Q: What's your plan for the remaining 70% of the project?

Point directly to the Pending list on the Implementation Status slide — sensor integration, dashboard-to-model linking, full helmet mounting, and field testing — as a clear phased roadmap targeted for Review 2.

Q: How do you know this is actually a real, significant problem worth solving?

This is backed by real accident and PPE-compliance data, not assumption — see Part C of this kit for the case study and survey evidence behind our Problem Statement.

Part C

Problem Validation — Case Study & Survey Evidence

Use this to answer "do you have data behind this?" — cite freely in Q&A

C1 · Scale of the Problem — Accident Statistics

- A study by NIT Surat and IIT Delhi found that of roughly 48,000 annual occupational deaths in India, **11,614 (24%)** occur in the construction sector alone — the single largest share of any industry.
- Karnataka's Building & Other Construction Workers Welfare Board recorded **223 construction worker deaths** between 2020–21 and 2025–26 in that state alone; experts note real numbers are likely higher since many cases go unreported, especially among unregistered laborers.
- A national analysis of construction deaths (2013–2016, 1,092 recorded deaths) found **60% resulted from falls from height**, about a quarter from structural or wall collapse, and 15% from electrocution — directly validating our fall-detection and PPE-monitoring focus.
- An academic study of Delhi Police FIR records (2016–2018, published in Injury Epidemiology) documented 929 construction-site accidents causing 1,217 injuries, of which **29% were fatal**. The same study notes construction workers are 3–4x more likely to die on the job than workers in other sectors — rising to roughly 6x in developing-world contexts.

Sources: ScienceDirect (Patel & Jha, 2016 study via NIT Surat/IIT Delhi); Karnataka Building & Other Construction Workers Welfare Board (via Business & Human Rights Centre / Deccan Herald); Insuropedia construction safety report; Injury Epidemiology journal, 2022.

C2 · PPE Non-Compliance — Survey Evidence

- Industry surveys show only about **60% of construction workers consistently wear PPE** despite well-documented hazards, and a separate study found **40.6% of construction workers fail to use PPE** altogether on the job.
- US Bureau of Labor Statistics data found **84% of workers who suffered head injuries** were not wearing head protection at the time of the accident.
- A safety-professional survey (ASSE / Kimberly-Clark) found **98% of respondents** had personally observed workers skipping required PPE on the job, with nearly a third saying it happens on numerous occasions — confirming this is a persistent, widespread behavior gap, not a rare lapse.
- Academic review studies place PPE compliance rates at just **50–75%** across construction settings, and estimate that improper or non-use of PPE contributes to **20–40% of construction injuries** overall.

Sources: EHS School — OSHA PPE Adherence Statistics (2025); EHS Today / Kimberly-Clark safety professional survey; HexArmor safety helmet statistics (BLS data); NCBI/PMC — PPE usage evaluation studies.

C3 · Existing Technology — Research Validation

- Peer-reviewed research independently validates YOLO-based PPE detection as a credible, production-worthy approach: one YOLOv11-based PPE detector achieved **94% precision, 90.1% recall, and 92.8% mAP@50** in real deployment testing, processing images in 2–3 seconds even on standard CPU hardware.
- A widely-cited benchmark study using a real construction-site dataset (CHV, 1,330 images) found YOLOv5x achieved **86.55% mAP** for helmet, vest, and person detection — confirming the YOLO family is well-suited to exactly the classes our project targets.

- Independent IoT research (published in IJERT and IJRIAS) describes smart-helmet designs built on **ESP32 + MPU6050 (fall) + gas sensor + DHT temperature/humidity sensor** — the same core hardware architecture we selected — reinforcing that our hardware design choices follow an established, peer-reviewed pattern rather than an untested one.

Sources: ResearchGate — "PPE detector: a YOLO-based architecture..."; NCBI/PMC — "Fast PPE Detection for Real Construction Sites" (CHV dataset study); IJERT — "Smart Workers Helmet"; IJRIAS — "GSM-based IoT Smart Safety Helmet".

How to Use This in the Review

If a jury member asks "is this a real problem" or "where's your evidence," don't just repeat the Problem Statement slide — cite one stat from C1 (scale), one from C2 (compliance gap), and one from C3 (this approach is already validated in research, we're extending it). That three-point answer covers scale, behavior, and technical credibility in under 30 seconds.

Reference List

- Patel, D.A. & Jha, K.N. (2016) — construction fatality share of India's occupational deaths, cited via ScienceDirect (sciencedirect.com/science/article/abs/pii/S0925753523000231).
- Karnataka Building & Other Construction Workers Welfare Board data — via Business & Human Rights Centre (business-humanrights.org) and Deccan Herald (deccanherald.com).
- Insuropedia — "Construction Site in India: A Big Risk For Workers" (securenow.in/insuropedia).
- Edwards, P. et al. (2022) — "They built this city: construction workers injured in Delhi, India," Injury Epidemiology, Vol. 9 (doi.org/10.1186/s40621-022-00388-4).
- EHS School — "OSHA PPE Adherence Statistics 2025" (ehsschool.com/blog/osha-ppe-adherence-statistics).
- EHS Today — "Workers Are Risking Injury By Not Wearing Safety Equipment" (ehstoday.com).
- HexArmor — "The Hard Truth About Safety Helmet Injuries and Statistics" (hexarmor.com).
- ResearchGate — "PPE detector: a YOLO-based architecture to detect PPE for construction sites".
- NCBI/PMC — "Fast PPE Detection for Real Construction Sites Using Deep Learning Approaches" (CHV dataset).
- IJERT — "Smart Workers Helmet" (ijert.org/smart-workers-helmet-ijertv15is070397).
- IJRIAS — "Design and Implementation of a GSM-based IoT Smart Safety Helmet for Construction Workers".